LONG-ONLY STRATEGY PROPOSAL

Screened Electrification Equipment

A rules-built long book of the profitable equipment and engineering suppliers to the electricity build-out — the picks-and-shovels of electrification, with the theme's ballast stripped out and three risk controls layered on.

Full backtest Jan 2019 – Aug 2026: **+348%** total return, **0.61** market beta, **–23%** max drawdown, Sharpe **1.05**. This note also documents an exhaustive search for a short offset — nine candidates — and concludes that none of them improves the book.

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1 What the book holds

The book is long roughly **27 companies that sell into the electricity build-out**: power equipment (transformers, switchgear, power electronics), grid-scale storage, and electrical engineering & construction. It holds *suppliers* — not utilities, not power generators. Representative names: Eaton, GE Vernova, Quanta Services, Vertiv, Hubbell, nVent, Fluence, Primoris.

The selection rule

A company enters the book only if it (a) sells to institutional buyers — utilities, hyperscalers, industrial customers, not consumers — and (b) meets at least two of three fundamental tests: **profitable today**, **valued on earnings** rather than on story, and **low dependence on subsidies or policy**. Names are held in **equal weight**, capped at 25% each. There is no cap-weighting, so the book does not concentrate in a few mega-caps the way a thematic index does.

Three risk controls

- **Volatility scaling.** The book automatically reduces its invested percentage when markets turn choppy and restores it when they calm, aiming to keep annual volatility near 20%.
- **Valuation trim.** After the book has run up well ahead of its own long-run trend and the broad market, it steps exposure down in stages — to about 80%, then 60% — taking risk off into euphoria.
- **A 15% reserve sleeve.** One-seventh of the portfolio is permanently held in gold and short-term US Treasuries instead of equities, as a shock absorber: a growth or credit scare pushes Treasuries up; a currency or geopolitical shock pushes gold up.

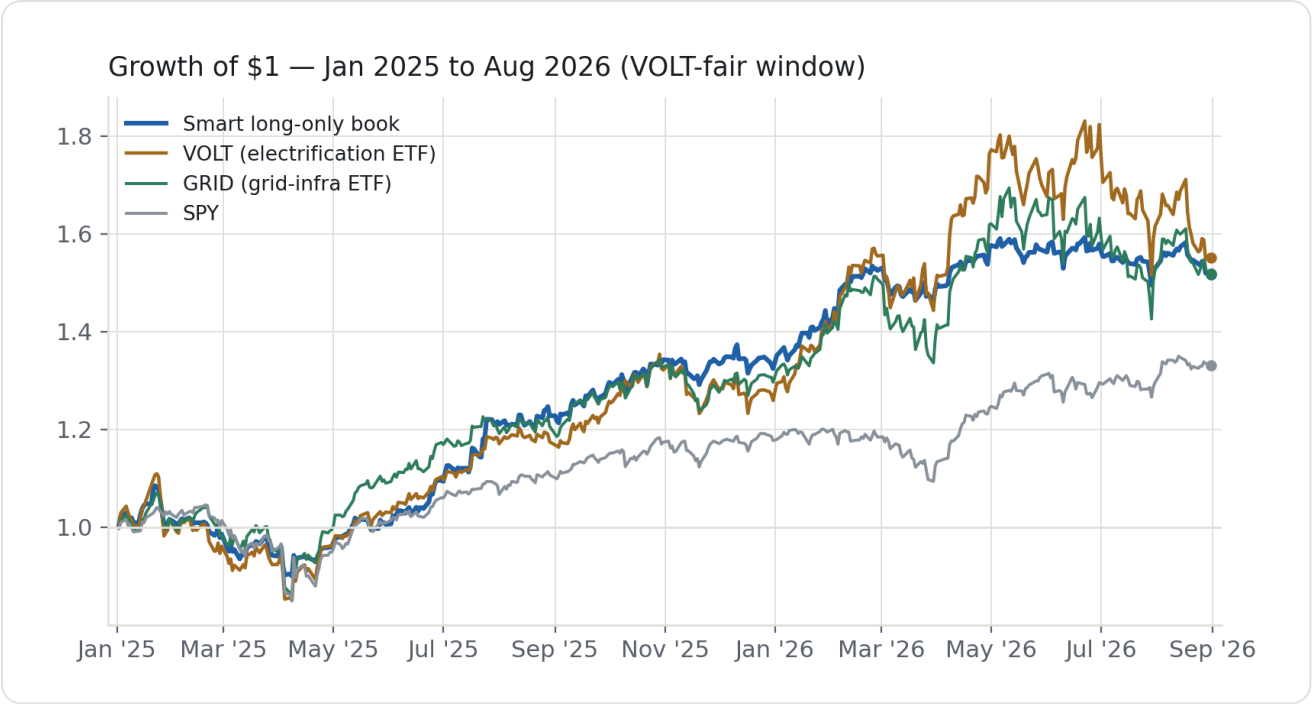
Is this just VOLT? In what way is it smarter?

It is the same theme. The book runs about **0.86 correlated** with VOLT, the electrification-theme ETF, and essentially all of its return is the theme. It is "picks and shovels." It is smarter than buying the ETF in four concrete ways:

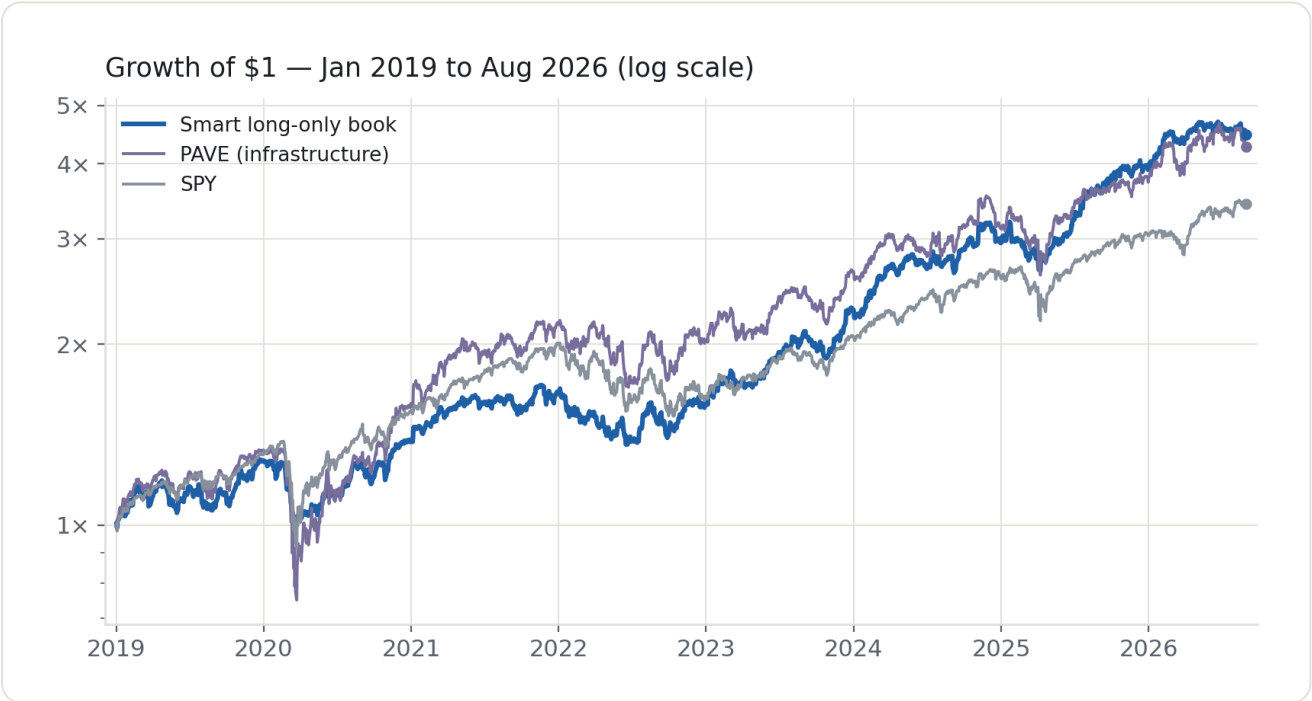
- **It strips the ballast.** No regulated utilities (~30% of VOLT), no pipelines or midstream (~10%), no commodity components (~13%). Pure equipment and EPC.
- **It screens on fundamentals.** Only profitable, earnings-valued, low-policy-dependence names — so it drops the pre-revenue, subsidy-reliant and story-stock exposure a thematic index sweeps in.
- **Equal weight, 25% cap** instead of cap weight — materially less single-name concentration.
- **Risk controls the ETF has none of** — the three above.

Over the only fair comparison window — January 2025 to August 2026, VOLT's live history — the book returns **Sharpe 1.49 vs VOLT's 1.05**, a **–17% max drawdown vs –23%**, at **63% of VOLT's volatility**, for slightly less raw return (CAGR 27.7% vs 30.5%). Over the full 2019–2026 backtest it earns Sharpe 1.05 at **0.61 market beta**: it captures the theme with roughly half the market sensitivity of the infrastructure ETFs.

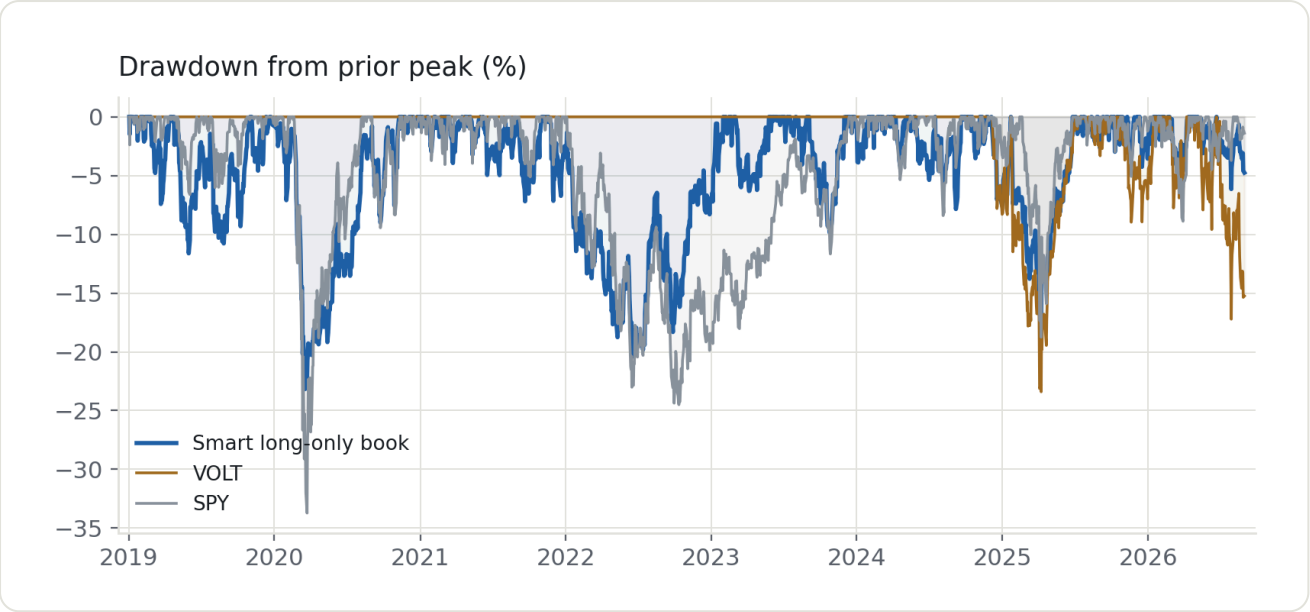
2 Performance



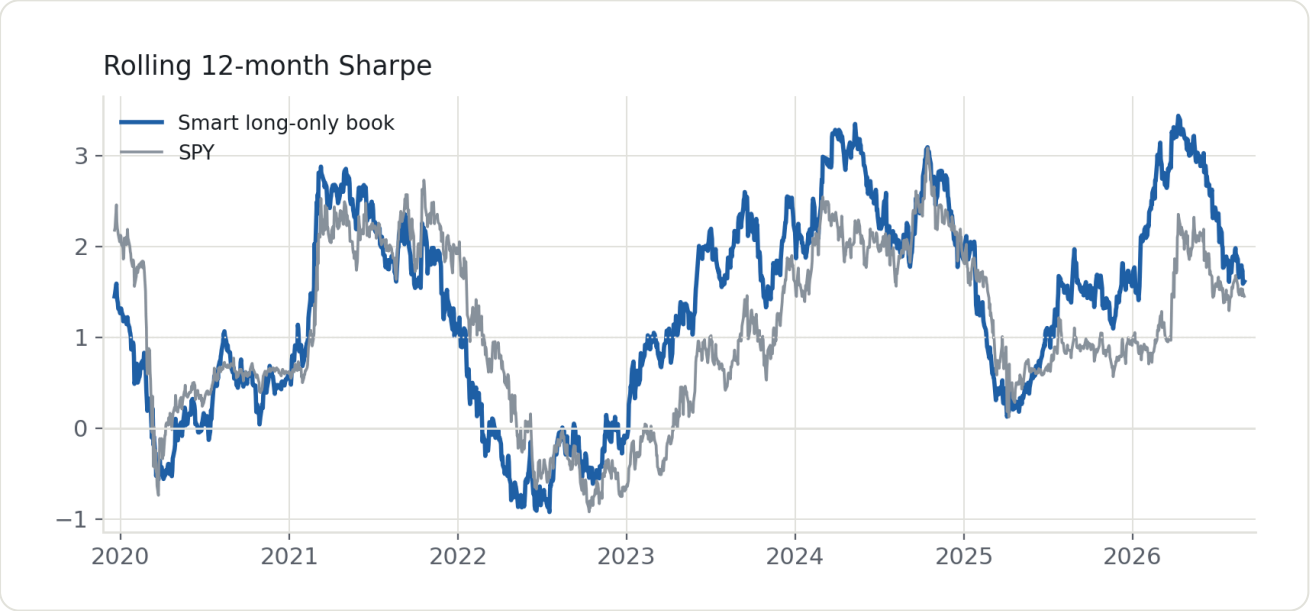
Fair-comparison window. The book keeps pace with the electrification ETFs through mid-2026 with a much smoother path — it does not spike to VOLT's 1.8× in May–July 2026, and it does not give it back.



Full backtest. The book trails through 2019–2022, then compounds through the electrification up-leg to finish level with the infrastructure ETF and ahead of the market — at lower volatility and a shallower worst drawdown than either.



Drawdown from prior peak. COVID: book −22% vs SPY −34%. 2022 rate shock: −20% vs −24%. The book’s worst point in the backtest is −23%.



Rolling 12-month Sharpe. Both go negative in 2022; the book leads the market for most of the record and through the 2023–2026 up-leg.

Headline statistics

STRATEGY	TOTAL RET	CAGR	ANN VOL	SHARPE	MAX DD	SPY B
Screened Electrification Equipment *(the book)*	+348%	+20.8%	16.3%	1.05	−23%	0.61
VOLT — electrification ETF	—[1]	—	25.5%	—	—	1.01
GRID — grid-infrastructure ETF	+397%	+23.3%	24.2%	0.81	−41%	1.07
PAVE — US infrastructure ETF	+327%	+20.9%	25.7%	0.67	−44%	1.12
SPY — S&P 500	+243%	+17.5%	19.3%	0.71	−34%	1.00
XLI — S&P industrials	+207%	+15.8%	21.5%	0.56	−42%	0.97
SPY + 15% GLD/IEF sleeve	+222%	+15.9%	16.3%	0.75	−29%	0.86

Full backtest Jan 2019 – Aug 2026 unless noted. Fair window = Jan 2025 – Aug 2026 (VOLT's live history). Sharpe = (CAGR − 3.76%) / annualised volatility. [1] VOLT listed in 2024, so its full-window statistics are not meaningful.

Behaviour in named stress episodes

EPISODE	BOOK	SPY	VOLT
COVID crash Feb–Apr 2020	−22%	−34%	—
Rate shock 2022	−20%	−24%	—
DeepSeek selloff Jan–Apr 2025	−17%	−19%	−23%
Tariff shock Apr 2025	−8%	−12%	−10%
2026 selloff YTD	−6%	−9%	−17%

3 Two-year split

Sharpe / total return / worst drawdown, computed separately for the 2023–2024 and 2025–2026 sub-periods. The *Why / what it is* column explains why each row is on the table.

ROW	WHY / WHAT IT IS	SH 23–24	RET 23–24	DD 23–24	SH 25–26	RET 25–26	DD 25–26
THE BOOK & ITS VARIANTS							
Screened Electrification Equipment *(the book)*	– Profitable equipment & engineering suppliers to the grid build-out, equal-weighted – Automatically shrinks exposure when markets get volatile or prices outrun their trend – Permanently holds 15% in gold and short-term Treasuries as a shock absorber	2.08	+86%	−10%	1.49	+52%	−17%
· raw book, no overlays	– The same names in equal weight, with none of the risk controls – Shown to isolate what the overlays cost in return and save in drawdown	1.87	+114%	−16%	1.27	+62%	−22%
· + built-in insurance short	– Adds a small conditional short in data-centre landlords (Digital Realty, Equinix) – Engages only while the 10-year real yield is rising; carried as paid tail insurance	1.81	+69%	−8%	1.66	+53%	−12%
· thematic universe + controls	– Same risk controls, but names chosen by "held by ≥2 electrification ETFs" – Broader, more index-like — drops the profitability screen	1.29	+61%	−14%	1.29	+49%	−19%
BOOK + A SHORT OFFSET							
book + short · XLI 0.25×	– Hold the book, short industrials at ~¼ of book size – Aim: cancel the book's general market and cyclical swings, keep the stock picks	2.13	+72%	−7%	1.43	+41%	−13%
book + short · China FXI 0.15×	– Hold the book, short Chinese large-caps at ~15% – A cyclical exposure that historically moves least in step with the book	1.97	+80%	−9%	1.42	+47%	−18%

ROW	WHY / WHAT IT IS	SH 23–24	RET 23–24	DD 23–24	SH 25–26	RET 25–26	DD 25–26
book + short · TAN solar 0.15×	– Hold the book, short solar and distributed-energy stocks at ~15% – The theorised natural offset — names that should gain from high power prices and green policy	2.81	+105%	−7%	1.35	+42%	−15%
book + short · merchant gen 0.15×	– Hold the book, short Vistra / NRG / Constellation / Talen at ~15% – The listed names most directly long the wholesale electricity price	1.32	+49%	−10%	1.58	+42%	−12%
BENCHMARKS							
VOLT	– Electrification-theme ETF, the closest off-the-shelf comparable – ~40% of it is utilities, pipelines and commodity components	—	—	—	1.02	+55%	−23%
GRID	– Grid-infrastructure ETF — T&D equipment plus utilities, weighted by size	0.83	+40%	−21%	1.04	+52%	−20%
PAVE	– Broad US infrastructure ETF (construction, materials, machinery) – The “just buy infrastructure” alternative	1.05	+54%	−13%	0.75	+37%	−23%
SPY	– The S&P 500 — the overall US stock market	1.65	+58%	−10%	0.85	+33%	−19%
XLI	– The S&P industrials sector — the book’s parent sector	0.95	+39%	−13%	0.85	+35%	−18%
SPY + 15% GLD/IEF sleeve	– The market with the book’s exact reserve sleeve bolted on – Isolates the sleeve from the stock selection	1.70	+52%	−9%	1.01	+34%	−16%

Sharpe = (CAGR − 3.76%) / annualised vol, same risk-free rate both periods. Max DD is the worst peak-to-trough *within* each period. VOLT has no 2023–24 history.

> **Read the shorts row-by-row.** XLI and FXI look strong in 2023–24 (Sharpe 2.1–2.0) — but that is the window the book beat its own sector by 2×, so shorting the laggard flatters the ratio. In 2025–26 every short cuts return and most cut Sharpe. TAN solar's 2.81 in 2023–24 is the solar crash, not a hedge: it adds nothing in 2025–26 and is an anti-hedge in the scenario the book most fears (see §6).

4 Systematic risks & macro bets

What the book is implicitly betting on, and how — or whether — the three overlays address each one.

RISK / MACRO BET	TREATMENT	WHAT IT IS, AND HOW THE STRATEGY HANDLES IT
Electrification theme de-rating	PARTIAL	AI-capex decelerates, data-centre moratoria spread, or grid capex plateaus and the whole theme re-prices down. Volatility scaling de-grosses on the way down and the valuation trim lowers exposure after run-ups; the sleeve is unaffected. Not eliminated — this is the primary disclosed risk.
Falling-rate, risk-on reversal	NOT HEDGED	Rates fall, the washed-out clean-energy complex rallies, and the theme de-rates in relative terms. By design there is no hedge: the Treasury leg of the sleeve gains as rates fall and gold is roughly neutral, but a solar short would *worsen* this — which is why there is none.
Broad equity drawdown	CUSHIONED	A market-wide selloff; the book carries 0.61 market beta. The 15% sleeve and volatility scaling both reduce the hit — the book fell −22% in COVID vs the market's −34%.
Rate shock / duration	MIXED	Yields spike fast (2022-style) and long-duration growth multiples compress. The Treasury leg of the sleeve *loses* in a rate spike; volatility scaling helps; the book still drew −20% in 2022 (vs the market's −24%).
Industrial / cyclical recession	LIMITED	Construction and capital-equipment orders slow. Grid capex is backlog-driven and partly regulated — more defensive than broad industrials — but not immune. No explicit hedge.
Policy / subsidy reversal	MITIGATED	IRA rollback or investment-tax-credit repeal. Mitigated by construction — the screen's "low policy dependence" test deliberately excludes the subsidy-reliant names.
Single-name concentration	BOUNDED	27 names, 25% cap; a blow-up in a top holding. Equal weight caps any one name near 4–8% of risk. It is not a 200-stock book — the concentration is deliberate and disclosed.
Input-cost inflation copper · electrical steel · labour	NOT A RISK	The book's real cost inputs spike. Tested directly: the book's beta to all three is ~0 — these suppliers pass costs through and had pricing power, and the book *outperformed* in the biggest cost-shock months.

5 When it can fail

- **A sustained rates-down, risk-on reversal.** The washed-out clean-energy complex rallies and the whole electrification theme de-rates. The overlays soften the path; they do not stop a fundamental repricing. This is the single biggest risk and it is not hedged.
- **A data-centre / AI-capex air pocket.** Hyperscaler capex decelerates, or local moratoria (Loudoun, Prince William, Central Ohio, North Texas) spread. Equipment backlogs cushion the first few quarters; a multi-year slowdown would not be cushioned.
- **The book is the theme.** ~0.86 correlated with VOLT, and essentially all of the return is thematic beta — enhanced beta, not alpha. If the theme is wrong, the book is wrong.
- **A 2022-style rate shock hurts twice** — growth multiples compress and the Treasury leg of the sleeve loses. The book still drew –20% then.
- **One macro cycle of evidence.** The backtest is 2019–2026, dominated by a single long electrification up-leg (2023–26). Regime diversity is limited.
- **Point-in-time universe not yet locked.** The screen's judgment columns carry some hindsight until a strict as-of-date version is built — a pre-live gate.

6 Short offsets: considered, not adopted

We tested **nine** short candidates as overlays on the book. **None improves risk-adjusted return.** The three most worth discussing are below; the rest are summarised after.

First, the book on its own

FOR

- Captures the electrification theme at ~0.6 market beta
- Best drawdown of any option here — −23% full, vs −34% to −44% for the ETFs and the market
- All the risk control is portable and cheap: no shorting, no borrow, ≤100% gross
- Beats every benchmark on Sharpe *and* drawdown in both sub-periods

AGAINST

- Still ~0.86 correlated with the theme — no explicit hedge for a theme de-rating
- Concentrated: 27 names, deliberately
- One-cycle backtest; point-in-time universe not yet locked
- The reserve sleeve's Treasury leg loses in a rate spike

BROAD-BETA HEDGE

Short industrials (XLI), 0.25×

Why we considered it. Of all the candidates it "looked good": it cuts the full-period max drawdown by ~7 points and, in 2023–24, lifted Sharpe to 2.1. The idea is to short a slice of the industrial sector to cancel the book's general market and cyclical swings and leave the stock selection.

SHARPE	TOTAL RETURN	MAX DD	FEARED SCENARIO
1.05 (book 1.05)	+238% (book +348%)	−16% (book −23%)	+2.2% (book +3.1%)

FOR

- Real drawdown reduction: −23% → −16% over the full backtest
- Sharpe is flat, not down — 1.05 either way

AGAINST

- Costs ~110 points of compound return (+348% → +238%)
- The 2023–24 Sharpe of 2.1 is regime luck — that is the window the book beat its sector by 2×
- A live bet that industrials keep lagging the book; ~125% gross, borrow cost, dividends paid on the short
- **A 25% cash / T-bill weight gives the same return and drawdown at a *higher* Sharpe (1.06)** — this is de-levering, not hedging

DIVERSIFIER

Short China large-cap (FXI), 0.15×

Why we considered it. It is the least-correlated global-industrial cyclical available as one liquid instrument, and it is the only candidate with a *positive* Sharpe delta (+0.02).

SHARPE	TOTAL RETURN	MAX DD	FEARED SCENARIO
1.06 (book 1.05)	+326% (book +348%)	−20% (book −23%)	+2.5% (book +3.1%)

FOR

- Cheapest of the lot — only ~22 points of full-period return given up
- Small drawdown help (~3 points); Sharpe nominally the best of any overlay

AGAINST

- +0.02 Sharpe is inside the noise — not a real improvement
- Adds a China-decoupling tail: a fat left tail whenever China rallies hard
- No exit rule; does not hedge the actual feared scenario (+2.5% vs the book's +3.1% in those months)

THEMATIC MIRROR — THE "NATURAL" OFFSET

Short solar / DER (TAN), 0.15×

Why we considered it. It is the textbook offset: short the names that should *benefit* from high power prices and green policy — the economic mirror of the long book.

SHARPE	TOTAL RETURN	MAX DD	FEARED SCENARIO
0.96 (book 1.05)	+259% (book +348%)	−20% (book −23%)	+0.3% (book +3.1%)

FOR

- In 2023–24 the combined Sharpe was 2.8 — superficially the best-looking result in the whole study

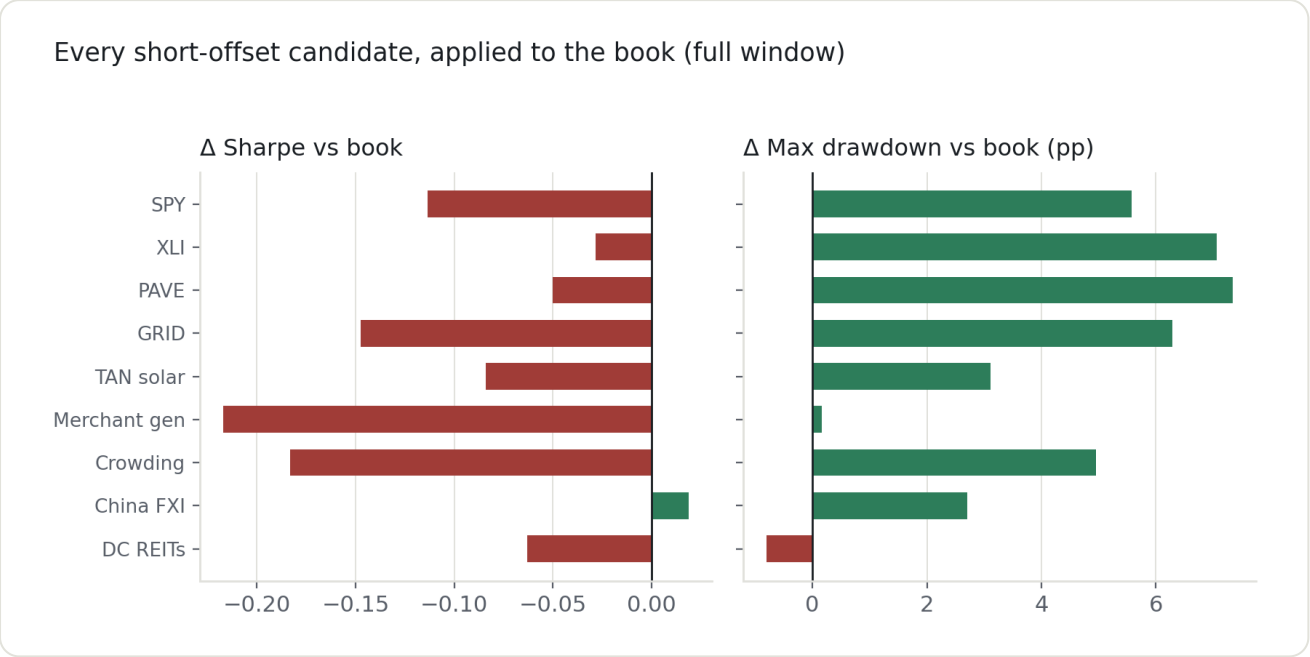
AGAINST

- That 2.8 *is* the 2023–24 solar crash, not a hedge — in 2025–26 it adds nothing (1.35 vs 1.49)
- **It is an anti-hedge in the feared scenario**: book +3.1% → +0.3% in those months, because "theme de-rates while clean energy rallies" is exactly when solar rips
- Prior tests at larger size took the book's Sharpe from 0.69 to −0.18 and its drawdown from −24% to −83%

The other six, briefly

- **Short SPY 0.25×** — the blunt beta hedge. Sharpe −0.11; same de-levering point as XLI, with more tracking error.
- **Short PAVE 0.25×** — Sharpe −0.05, drawdown −7 points; a slightly worse XLI.
- **Short GRID 0.25×** — the purest theme-relative construction. Sharpe −0.15; you are shorting your own thesis.
- **Short the merchant generators (VST/NRG/CEG/TLN) 0.15×** — the names literally long the power price. Sharpe −0.22 *and* drawdown slightly *worse*: they run +0.6 correlated with the book, so this shorts the book's own beta.
- **Short a crowded-adjacents basket 0.15×** — Sharpe −0.18; bleeds borrow and carry.

- **Short data-centre REITs (DLR/EQIX) 0.10×** — the runner-up "insurance" leg. Sharpe −0.06, drawdown unchanged; as a *conditional* short (only while real yields rise) it is the one variant that helped in 2025–26, at a cost in 2023–24.



Every candidate, full backtest. Left: all but one *reduce* the book's Sharpe. Right: most trim drawdown by 1–7 points — the exception, shorting merchant generators, makes it worse.

Verdict — no short offset

No equity short improves the book on a risk-adjusted basis. The candidates that reduce drawdown do it by **shrinking net exposure** — and a cash or T-bill weight does that more cheaply, at a higher Sharpe, with no borrow, no dividends paid on a short, and half the gross exposure.

FULL BACKTEST	SHARPE	TOTAL RET	MAX DD
Book, 100%	1.05	+348%	−23%
Book + short XLI 0.25×	1.02	+238%	−16%
75% book + 25% short-term Treasuries	1.06	+240%	−18%

If the mandate requires a lower drawdown, the sanctioned dial is **75–85% book + short-term Treasuries**, not a short. Otherwise drawdown control lives in the three built-in overlays — volatility scaling, the valuation trim, and the 15% reserve sleeve.

METHOD & CAVEATS

Window. Backtest runs Jun 2017 – Aug 2026; all headline statistics and figures use Jan 2019 onward, before which the screen holds fewer than eight listed names and the series is a near-empty warm-up.

Sharpe. (CAGR – 3.76%) / annualised volatility; 3.76% ≈ the mean 3-month US T-bill over the backtest. The same risk-free rate is used in both sub-periods, which marginally flatters the 2023–24 figures.

Construction. Book returns are the `screen +val+sleeve` variant of the `electrification_strategy` package. Overlays and the reserve sleeve are applied as daily-rebalanced constant weights. Short overlays subtract `w ×` the instrument's daily return; broad-index candidates at `w = 0.25`, thematic baskets at `w = 0.15`, data-centre REITs at `w = 0.10`.

Feared scenario. The 15 calendar months in which solar (TAN) most outran the market — a proxy for "the theme de-rates while clean energy rallies."

Data. Prices from Yahoo Finance adjusted close. VOLT listed in 2024, so its full-window statistics are omitted and its fair window is Jan 2025 – Aug 2026.

Simulated historical performance. Past simulated results are not indicative of future performance. This document is research, not investment advice.